

Denish Hirpara

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SUMMARY

Machine Learning Engineer with 3 years of experience in Generative AI, RAG systems, fine-tuning Large Language Models (GPT models), prompt engineering, building evaluation pipelines on the LangSmith, ML developing object detection pipelines and utilizing cloud platform service like Vertex AI for scalable training.

EDUCATION

Dhirubhai Ambani University

B.Tech. (Honours) in ICT with minor in Computational Science, CGPA : 8.24

July 2019 – May 2023

Gandhinagar, Gujarat

EXPERIENCE

Machine Learning Engineer

Infocusp Innovations LLP

July 2023 – Present

Ahmedabad, India

Dscout - Insights Automation Platform

- Optimized workflows for multimodal data (text, image, video), enabling automated theme extraction, summarization, question validation, and notable moment detection across research datasets.
- Demonstrated expertise in natural language processing (NLP), prompt engineering for large language models, OpenAI API integration, and multimodal data analysis across text, images, and video.
- Fine-tuned LLMs to boost accuracy by **25%**, improving overall insight precision.
- Built an offline evaluation pipeline and integrated LangSmith tracing to monitor prompt performance, track model behavior, and systematically improve output quality over time.
- Reduced manual research analysis time by **60%** through automated insight extraction, enabling UX teams to focus on higher level strategy.

UX Researcher Assistant

- Built a retrieval-augmented generation (RAG) pipeline to power AI workflows, enabling researchers to experiment with prompts, retrieve relevant information efficiently, and extract actionable insights.
- Developed a RAG-powered chatbot with interactive UI for researchers, including a prompt versioning system to iterate and improve chatbot behavior over time.
- Minimized manual effort in analyzing user feedback by automating insight extraction, allowing researchers to focus more on interpretation and strategy.

Data Processing and API Cost Estimation System

- Processed structured and unstructured data from pre-collected URLs to extract actionable insights and support downstream analytics workflows.
- Developed an API cost estimation tool to forecast, monitor, and optimize usage costs across multiple model calls and pipeline stages.

Software Engineering Intern (ML)

Infocusp Innovations LLP

January 2023 – June 2023

Ahmedabad, India

Object Detection Pipeline

- Developed and implemented an object detection pipeline using TensorFlow Model library, optimizing training and inference processes.
- Designed a scalable pipeline for training models on custom datasets, enhancing accuracy and efficiency in object detection tasks.
- Utilized cloud platform service such as VertexAI to train models at scale for enhanced performance and efficiency and deployed models with integrated endpoints for efficient inference in production environments.

SKILLS

Languages: Python, C++

Data Processing & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Machine Learning & AI: TensorFlow, PyTorch, Keras, NLTK, Scikit-learn, Hugging Face, OpenCV

LLM: Prompt Engineering, Fine-tuning, OpenAI, Gemini, LangChain, LangGraph, LangSmith, RAG

Cloud Platforms & Tools: AWS(SageMaker, EC2), GCP (Vertex AI), Docker, Git, GitHub